

W2 Baseline Report: MLOps Pipeline for Medical Model Deployment

ChestMNIST — Data Exploration & Baseline Models

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Abstract—This report documents Week 2 deliverables for the MLOps Pipeline for Medical Model Deployment project. We present the data acquisition procedure, exploratory data analysis (EDA) of the ChestMNIST dataset, and the implementation and evaluation of two baseline models. Both baselines use a frozen ImageNet backbone with a trainable linear classification head (linear probing): MobileNetV2 and ResNet18. On the 14-class multi-label chest X-ray classification task, MobileNetV2 achieves a test macro-AUC of 0.6852 and ResNet18 achieves 0.6865, establishing reproducible reference points for full fine-tuning experiments in Week 3.

I. Dataset

A. Acquisition

ChestMNIST is part of the MedMNIST v2 benchmark collection [1], derived from the NIH ChestX-ray14 dataset. It requires no credentialing or manual download and is accessed programmatically via the medmnist Python package:

```
from medmnist import ChestMNIST
ds = ChestMNIST(split="train", download=True)
```

The dataset is downloaded on first call (~ 85 MB) and cached locally. All splits are loaded lazily — no upfront resizing is performed, avoiding the ~ 15 GB memory footprint that would result from pre-resizing all images to 224×224 in RAM.

B. Structure

ChestMNIST contains 112,120 frontal-view chest X-ray images, each labelled with up to 14 thoracic pathology classes in a multi-label binary format. The 14 classes are: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia. The official train/val/test split is used throughout, with all random operations fixed at seed 42.

Table I summarises the number of images and batches per split at batch size 32.

TABLE I
Dataset Split Sizes

Split	Images	Batches (bs=32)
Train	78,468	2,452
Validation	11,219	351
Test	22,433	701
Total	112,120	—

Raw images are 28×28 grayscale (uint8), as illustrated in Fig. 1. For pretrained backbone compatibility, each image is resized to 224×224 and replicated to 3 channels at load time via `torchvision.transforms`. Labels are cast to float32 (BCEWithLogitsLoss requires float targets).



Fig. 1. Sample training images with their ground-truth labels. Images are 28×28 grayscale; approximately 53% carry no pathology label (“No disease detected”).

C. Data Preprocessing Pipeline

All preprocessing is implemented in `torchvision.transforms` and applied on-the-fly at batch load time to avoid memory overhead. The pipeline differs between training and evaluation splits:

Training transform:

- 1) `Resize((224, 224))` — bilinear upscaling from 28×28 .
- 2) `RandomHorizontalFlip()` — data augmentation ($p=0.5$).
- 3) `ToTensor()` — scales pixel values from $[0, 255]$ to $[0.0, 1.0]$.
- 4) `Normalize(mean=0.5, std=0.5)` — maps to $[-1, 1]$.

Validation / Test transform:

- 1) `Resize((224, 224))` — bilinear upscaling.

- 2) ToTensor() — scales to $[0.0, 1.0]$.
- 3) Normalize(mean=0.5, std=0.5) — maps to $[-1, 1]$.

A GrayscaleTo3Ch adapter is applied inside the backbone wrapper to replicate the single grayscale channel to 3 channels, satisfying the input contract of ImageNet-pretrained models. Dataloaders use num_workers=0 and a seeded torch.Generator for fully reproducible batch ordering.

Fig. 2 compares the native 28×28 resolution against the 224×224 bilinear upscaling used at inference. Large-scale findings (effusions, cardiomegaly) remain visually identifiable after upscaling, while fine-grained findings (nodules, subtle infiltrates) are not recoverable at native resolution.

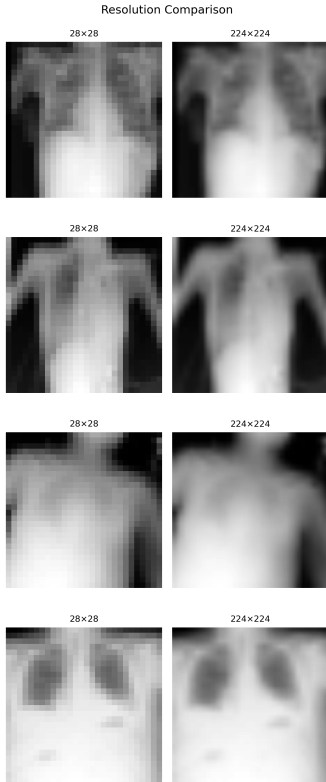


Fig. 2. Resolution comparison: native 28×28 (left) vs. bilinear upscaling to 224×224 (right) for four training examples. Large structural patterns are preserved; fine-grained details are not recoverable.

II. Exploratory Data Analysis

A. Data Integrity

A preliminary integrity check confirmed zero NaN values in both images and labels, pixel values strictly in $[0, 255]$, binary labels $\{0, 1\}$, and no blank images (no all-zero pixel vectors). No cleaning or imputation was required.

B. Class Distribution and Imbalance

ChestMNIST exhibits severe class imbalance across all 14 pathologies. Fig. 3 shows the absolute positive sample count per class in the training set. Infiltration (13,914 positives) and Atelectasis (7,996 positives) are the most prevalent

pathologies; Hernia is the rarest with only 144 positive training samples.

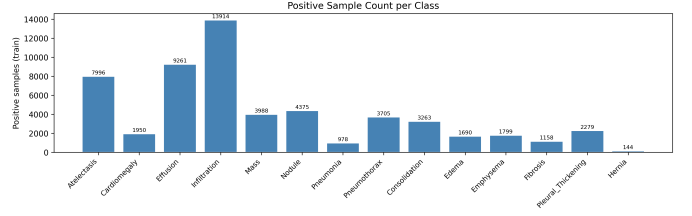


Fig. 3. Positive sample count per class in the training split. Infiltration and Atelectasis dominate; Hernia has only 144 positive examples out of 78,468 training images.

Fig. 4 quantifies imbalance severity as the negative-to-positive ratio per class on a log scale. All 14 classes exceed the 10:1 imbalance threshold. Hernia reaches a ratio of approximately 543:1, Pneumonia 79:1, and Fibrosis 67:1. Even the least imbalanced class (Infiltration) has a ratio near 5:1. Approximately 53% of training images carry no label whatsoever.



Fig. 4. Class imbalance severity (neg/pos ratio, log scale, sorted descending). Every class exceeds the 10:1 threshold (dashed line), making class-weighted loss mandatory.

C. Multi-label Density

Fig. 5 shows the distribution of the number of positive labels per training image. Over 54% of images have zero positive labels, 28% have exactly one, and 13% have two. Fewer than 2% carry three or more labels. The mean label count per image is 0.44, confirming the dataset’s sparse multi-label structure. The maximum observed label count on a single image is 9, though such cases are extremely rare.

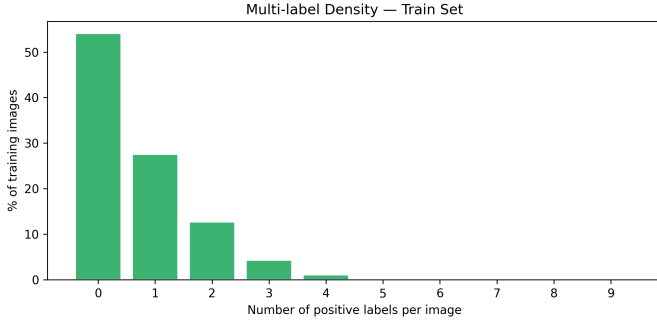


Fig. 5. Multi-label density in the training set. Over 54% of images have no positive label; the distribution drops sharply beyond 2 labels per image.

D. Label Co-occurrence

Fig. 6 shows the Jaccard co-occurrence matrix computed on the training set. Off-diagonal values are uniformly low (all below 0.16), confirming that pathologies tend to appear independently. The strongest co-occurrence pair is Effusion–Atelectasis (Jaccard = 0.15), followed by Effusion–Infiltration (0.14) and Atelectasis–Infiltration (0.12). Hernia is fully independent of all other pathologies (all Jaccard values = 0.00).

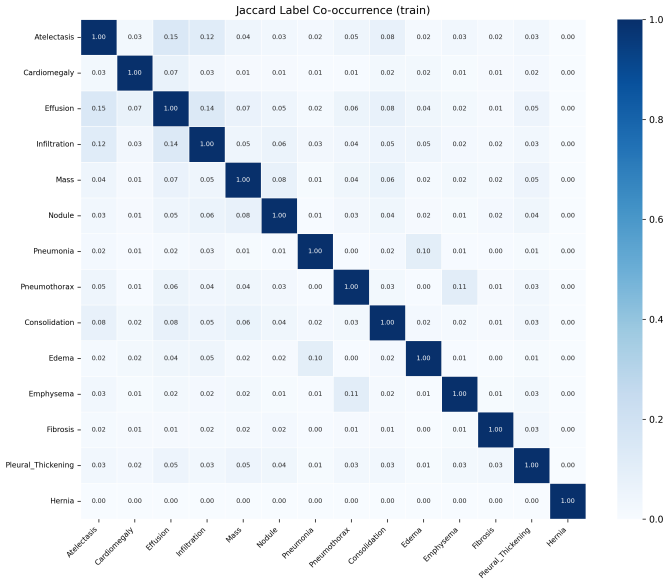


Fig. 6. Jaccard label co-occurrence matrix (train set). Values are low across all pairs, indicating weak label dependency. Effusion–Atelectasis is the strongest co-occurring pair (0.15).

The Pearson label correlation matrix (Fig. 7) is consistent with the Jaccard analysis. The highest off-diagonal Pearson correlations are Pneumonia–Edema (0.18), Emphysema–Pneumothorax (0.18), and Effusion–Atelectasis (0.17). All correlations remain below 0.20, confirming the absence of strongly redundant label pairs.

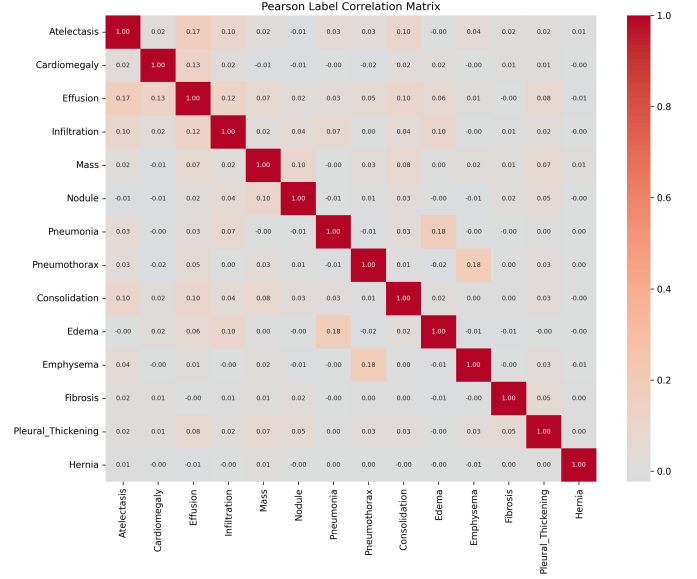


Fig. 7. Pearson label correlation matrix (train set). No pair exceeds 0.20 correlation, consistent with the sparse multi-label structure observed in the Jaccard analysis.

E. Pixel Statistics

Fig. 8 shows the pixel intensity histogram computed on a 5,000-image random sample. The distribution is roughly bimodal: a sharp peak near zero (dark background) and a broad bell around the midrange (lung tissue). The mean pixel intensity is 126.0 (uint8 scale), and the standard deviation is approximately 55. After normalization (mean 0.5, std 0.5), values map to $[-1, 1]$.

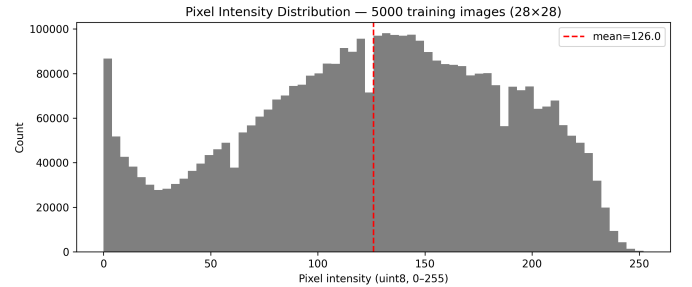


Fig. 8. Pixel intensity distribution over 5,000 training images (28×28 , uint8). The mean intensity is 126.0 (red dashed line). The bimodal shape reflects the dark background and brighter lung regions.

F. Per-split Consistency

Fig. 9 shows label prevalence across train, validation, and test splits. All 14 classes are consistent within $\pm 1\%$ across splits, confirming that the official MedMNIST split is well stratified. No split-specific resampling is needed.

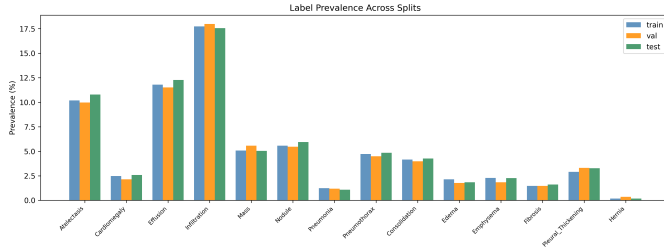


Fig. 9. Label prevalence (%) across train, validation, and test splits. All classes are consistent within $\pm 1\%$, confirming well-stratified official splits.

G. Loss Function Motivation

A preliminary experiment compared standard BCEWithLogitsLoss against the class-weighted variant (pos_weight) on a small CNN trained for 120 steps. Without pos_weight, the model collapses to predicting all-negative (macro-F1 = 0.000, consistent with the 53% no-finding majority). With pos_weight set to the per-class neg/pos ratio, macro-F1 rises to ~ 0.08 on the same architecture. The pos_weight tensor is computed as:

$$w_c = \frac{N_{\text{neg},c}}{N_{\text{pos},c}}, \quad c = 1, \dots, 14 \quad (1)$$

and passed directly to all subsequent training runs. This weighting is mandatory, not optional, for this dataset.

III. Baseline Models

A. Architecture

Both baselines follow a linear probing strategy: the backbone is loaded with ImageNet weights and fully frozen; only the final classification head is trained. A GrayscaleTo3Ch adapter replicates the single input channel to 3 channels before the backbone. Table II compares the two architectures.

TABLE II
Baseline Architecture Comparison

Property	MobileNetV2	ResNet18
Backbone params	3.5M	11.2M
Trainable params	17,934	513,024
Head	Linear(1280,14)	Linear(512,14)
Strategy	Linear probe	Linear probe

B. Training Setup

Both models are trained with the hyperparameters listed in Table III. The learning rate of 10^{-3} is standard for Adam on linear heads; backbone parameters receive no gradients. All experiments are tracked with MLflow (local mlruns/ store), logging per-epoch train loss, validation loss, macro-AUC, macro-F1, and per-class AUC. Best checkpoints (by validation macro-AUC) are saved to models/.

TABLE III
Training Hyperparameters (both models)

Parameter	Value
Loss	BCEWithLogitsLoss + pos_weight
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	32
Epochs	5
Input size	$224 \times 224 \times 3$
Seed	42

IV. Results

A. Test Set Performance

Table IV reports the primary test set metrics for both linear probes. Both models achieve macro-AUC near 0.686, with ResNet18 marginally ahead by 0.001. The low macro-F1 scores reflect the poorly-calibrated 0.5 threshold under severe class imbalance, not a failure of discriminative ability.

TABLE IV
Baseline Test Set Results

Metric	MobileNetV2	ResNet18
Test Macro-AUC	0.6852	0.6865
Test Macro-F1	0.1333	0.1400
Best Val AUC	0.6904	0.6990
Trainable params	17,934	513,024

B. Training Dynamics

Both models show the same pattern: train loss decreases steadily while validation loss diverges after epoch 2–3. The AUC curve is flat from epoch 1 onwards on both models, confirming the linear head reaches its representational ceiling within the first epoch. This behavior is expected and inherent to linear probing.

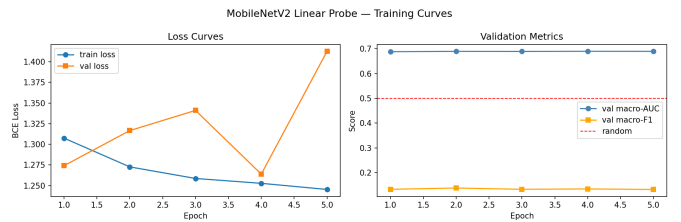


Fig. 10. MobileNetV2 training curves. Val loss diverges from epoch 3, confirming the linear head saturates quickly.

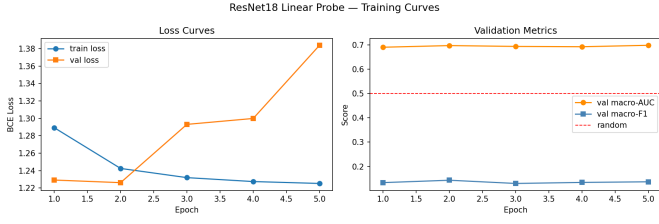


Fig. 11. ResNet18 training curves. Same divergence pattern from epoch 2, with faster saturation due to the smaller head (512 vs. 1280 features).

C. Per-class AUC

Per-class AUC is shown in Figs. 12 and 13. High-AUC classes correspond to pathologies with large-scale radiographic patterns (Edema, Effusion, Cardiomegaly); low-AUC classes require fine-grained texture features unavailable in frozen ImageNet representations (Nodule, Pneumonia). The ranking is nearly identical across both backbones.

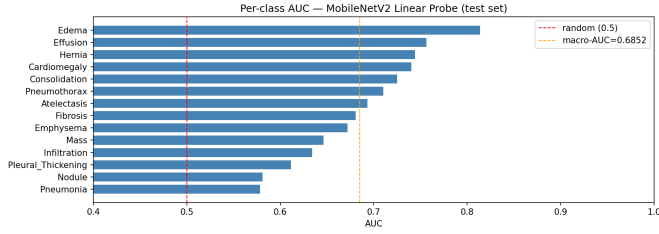


Fig. 12. Per-class AUC — MobileNetV2 linear probe (test set). Edema (0.81) and Effusion (0.76) are well above the macro average; Nodule (0.58) and Pneumonia (0.58) are near random.

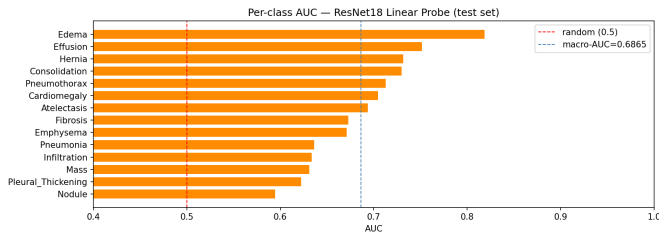


Fig. 13. Per-class AUC — ResNet18 linear probe (test set). Rankings are near-identical to MobileNetV2, confirming the per-class pattern is backbone-independent at this stage.

D. Head-to-Head Comparison

MobileNetV2 and ResNet18 are within 0.02 AUC on every individual class under linear probing, with no backbone systematically dominating across pathologies. Given near-identical accuracy, MobileNetV2 is operationally preferable due to its $28\times$ fewer trainable parameters.

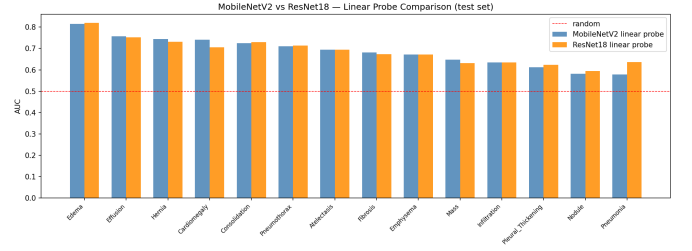


Fig. 14. MobileNetV2 vs. ResNet18 per-class AUC head-to-head. Both backbones are within 0.02 AUC on every class under linear probing.

E. Comparison with SOTA

The ≈ 0.08 macro-AUC gap to the MedMNIST full fine-tuning benchmark is consistent across both backbones. It is largest for Hernia and Emphysema (texture-dependent classes) and smallest for Consolidation and Atelectasis (large-scale structural patterns).

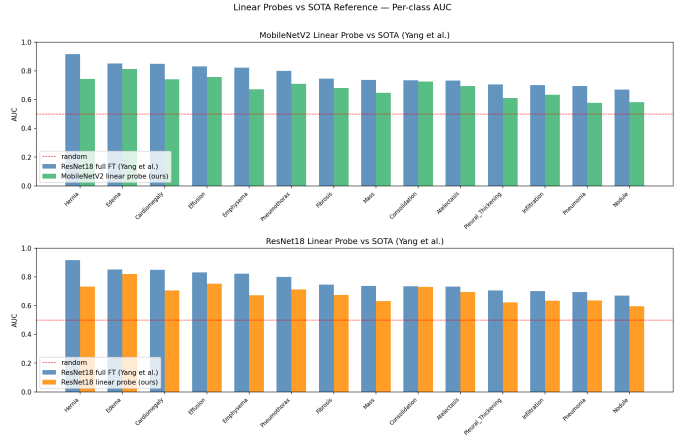


Fig. 15. Linear probes vs. ResNet18 full fine-tuning (Yang et al. [1]). Top: MobileNetV2 probe vs. SOTA. Bottom: ResNet18 probe vs. SOTA. The gap is consistent across both backbones (≈ 0.08 macro-AUC).

V. Interpretation

Both backbones are equivalent under linear probing. MobileNetV2 (0.685) and ResNet18 (0.687) achieve nearly identical macro-AUC, with per-class differences below 0.02 on every class. At this stage, the backbone choice does not matter for accuracy — both are bottlenecked by the frozen feature constraint.

AUC is meaningful; F1 is not yet. Macro-AUC well above 0.5 on both models confirms frozen ImageNet features transfer non-trivially to chest X-ray classification. Macro-F1 of ~ 0.13 is uninformative: the fixed 0.5 threshold is poorly calibrated for severe multi-label imbalance and does not reflect discriminative ability.

Per-class performance correlates with visual scale. Pathologies with large-scale distinctive patterns — Edema

(0.81–0.82), Effusion (0.75–0.76), Cardiomegaly (0.71–0.74) — are well-detected by frozen features. Subtle small-scale findings — Nodule (0.57–0.59), Pleural Thickening (0.61–0.62) — are near random on both backbones, as they require task-specific feature learning.

MobileNetV2 is operationally preferable. Near-identical accuracy with $28\times$ fewer trainable parameters (17K vs. 513K) and a lighter backbone (3.5M vs. 11.2M total params) makes MobileNetV2 the preferred candidate for the inference latency benchmarks in Week 3.

Gap to SOTA is ~ 0.08 macro-AUC on both backbones. This gap is consistent and expected for linear probing. It is largest for Hernia and Emphysema (texture-dependent, require deep feature adaptation) and smallest for Consolidation and Atelectasis (large-scale patterns, transfer well).

VI. Week 3 Plan

- 1) Full fine-tuning of both backbones ($\text{lr}=10^{-4}$, all layers unfrozen) to close the ~ 0.08 AUC gap to SOTA.
- 2) Inference latency benchmark: MobileNetV2 vs. ResNet18 — ms/image and ms/batch on CPU and GPU.
- 3) Per-class threshold tuning on the validation set to produce meaningful F1 scores.
- 4) FastAPI serving of the best checkpoint with a POST /predict endpoint.
- 5) Drift simulation and detection on the test set.

Reproducibility

All code is in notebooks/W2_baseline_experiments.ipynb. Environment: `pip install -r requirements.txt`. MLflow UI: `mlflow ui -port 5000` from the project root.

References

- [1] J. Yang et al., “MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification,” *Scientific Data*, vol. 10, no. 1, 2023.
- [2] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” *CVPR*, 2018.
- [3] M. Raghu, C. Zhang, J. Kleinberg, and S. Bengio, “Transfusion: Understanding Transfer Learning for Medical Imaging,” *NeurIPS*, vol. 32, 2019.
- [4] M. Zaharia et al., “Accelerating the Machine Learning Lifecycle with MLflow,” *IEEE Data Engineering Bulletin*, vol. 41, no. 4, pp. 39–45, 2018.